STPM 2017 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper. In Question 20(c) the formula is $p = \frac{1}{3}\rho\langle c^2\rangle$, as amended by the erratum issued with the paper.

Section A [15 marks]

1. The velocity, v, of a standing wave produced by a string is related to its tension, F, length, l, and mass, m. Which equation shows a possible relation between v, F, l and m if k is a dimensionless quantity?

A $v = k\sqrt{(\frac{Fl}{m})}$ B $v = kl\sqrt{(\frac{F}{m})}$ C $v = k\sqrt{(\frac{Fm}{l})}$ D $v = k\frac{\sqrt{(Fl)}}{m}$

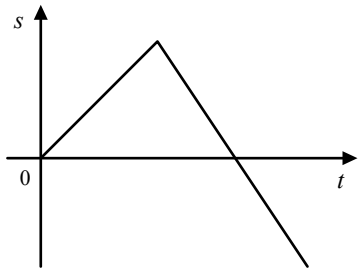
Answer: A

$$[\frac{Fl}{m}] = \frac{(MLT^{-2})(L)}{M} = L^2T^{-2} \Longrightarrow [\sqrt{(\frac{Fl}{m})}] = LT^{-1} = [v]$$

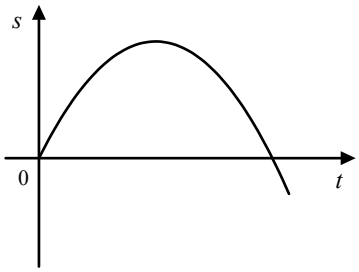
Explanation: An equation is only possible if both sides have the same dimensions. Check each option: only $\sqrt{(\frac{Fl}{m})}$ has the dimensions of velocity.

2. A stone is projected vertically upwards from the top of a building. Which graph shows the variation of the displacement, s, of the stone with time, t?

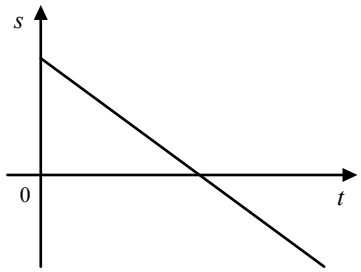
A



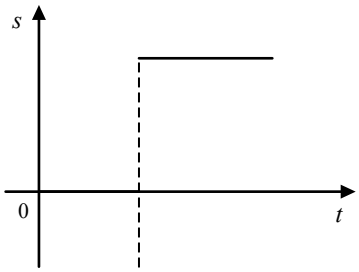
B



C



D



Answer: B

Taking the top of the building as $s = 0$ and upwards as positive: $s = ut - \frac{1}{2}gt^2$, a parabola that rises, reaches a maximum and then falls below zero (below the roof level).

Explanation: With constant acceleration, displacement is a quadratic function of time, so the graph is a curve, not straight lines (A) or a straight line (C). The stone ends up below its starting point, so s becomes negative.

3. Sphere A and sphere B collide head-on elastically. Which statement is true of the collision?

A The loss of momentum by sphere A is equal to the loss of momentum by sphere B.
B After the collision, the momentum of sphere A is equal to the momentum of sphere B.
C The loss of kinetic energy by sphere A is equal to the gain in kinetic energy by sphere B.
D After the collision, the kinetic energy of sphere A is equal to the kinetic energy of sphere B.

Answer: C

In an elastic collision the total kinetic energy is conserved, so the kinetic energy lost by A equals the kinetic energy gained by B.

Explanation: Momentum is conserved too: the momentum lost by A equals the momentum **gained** by B (not lost, A). The final momenta and kinetic energies need not be equal (B, D).

4. Which situation will **not** happen based on Newton's first law of motion?

A A force is required to change the state of motion of an object.
B When a push is removed from a moving object, the object stops moving.
C When an object is at rest, the nett force acting on the object is equal to zero.
D When an object moves with a constant velocity, the nett force acting on the object is equal to zero.

Answer: B

Newton's first law: an object continues in its state of rest or uniform motion unless a resultant force acts on it. A moving object does not stop just because the push is removed; it stops only if a force such as friction acts.

Explanation: Statements A, C and D all agree with the first law. Statement B is the old (Aristotelian) idea that a force is needed to keep an object moving.

5. A pendulum bob which is attached to a string of length ℓ is released at a position making an angle θ with respect to the vertical. What is the speed of the bob when it is at the lowest position?

A $\sqrt{(2g\ell)}\cos\theta$
B $\sqrt{(2g\ell)}\,(1-\cos\theta)$
C $\sqrt{(2g\ell\cos\theta)}$
D $\sqrt{(2g\ell(1-\cos\theta))}$

Answer: D

$$\text{Height fallen} = \ell - \ell\cos\theta = \ell(1-\cos\theta)$$

$$\frac{1}{2}mv^2 = mg\ell(1-\cos\theta) \Longrightarrow v = \sqrt{(2g\ell(1-\cos\theta))}$$

Explanation: Use conservation of energy. The key step is the vertical height through which the bob falls, found from the geometry of the string.

6. An object rests on a disc which rotates at a constant speed. Which statement is true about the object?

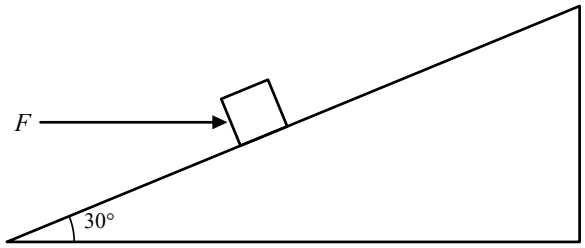
A The linear momentum is constant.
B The weight of the object acts as the centripetal force.
C The magnitude of the centripetal acceleration is constant.
D The angular and linear velocities are constant.

Answer: C

Uniform circular motion: the speed and radius are constant, so the centripetal acceleration v^2/r has a constant magnitude (its direction keeps changing).

Explanation: The linear momentum and linear velocity change direction (A and D wrong). The centripetal force is provided by static friction, not the weight, which is vertical (B wrong).

7. A block which weighs 100 N is placed on a smooth inclined plane at an inclination of 30° with the horizontal as shown in the diagram below.



What is the magnitude of the horizontal force, F, acting on the block so that it is in static equilibrium?

A 50 N B 58 N C 87 N D 100 N

Answer: B

Along the smooth plane: $F \cos 30° = W \sin 30°$

$F = W \tan 30° = 100 \tan 30° = 58$ N

Explanation: Resolve the forces parallel to the plane: the component of F up the plane balances the component of the weight down the plane. There is no friction.

8. Cylinder X contains an ideal gas at pressure 40 Pa, a temperature of 200 K and has N molecules. The volume of cylinder Y is half the volume of cylinder X. Cylinder Y contains $8N$ molecules at a temperature of 100 K. What is the pressure of the ideal gas in cylinder Y?

A 40 Pa B 80 Pa C 160 Pa D 320 Pa

Answer: D

$$p = \frac{NkT}{V} \Rightarrow \frac{p_Y}{p_X} = \frac{8N}{N} \times \frac{100}{200} \times \frac{V}{V/2} = 8$$

$$p_Y = 8 \times 40 = 320 \text{ Pa}$$

Explanation: Form a ratio using $pV = NkT$: the pressure is proportional to the number of molecules and the temperature, and inversely proportional to the volume.

9. A wire of length 5.00 m and a cross-sectional area of 1.20×10^{-4} m^2 is stretched to an extension of 1.84 mm. If the stress exerted on the wire is 1.23×10^8 N m^{-2}, what is the strain energy which is stored in the wire?

A 8.33 J B 13.6 J C 27.2 J D 67.9 J

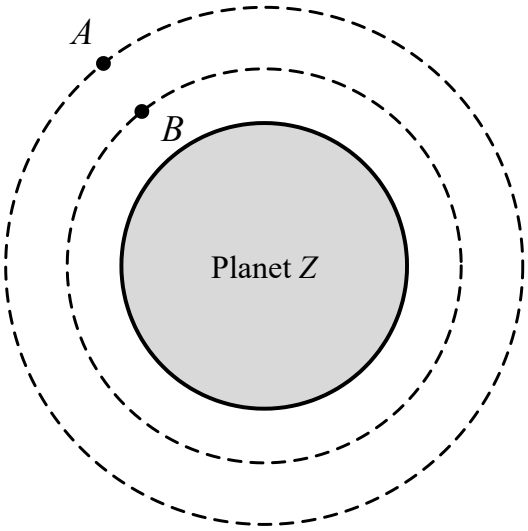
Answer: B

$$F = \sigma A = (1.23 \times 10^8)(1.20 \times 10^{-4}) = 1.476 \times 10^4 \text{ N}$$

$$U = \frac{1}{2}Fe = \frac{1}{2}(1.476 \times 10^4)(1.84 \times 10^{-3}) = 13.6 \text{ J}$$

Explanation: The strain energy is $\frac{1}{2}Fe$ (area under the force–extension graph). Find the force from stress × area and use the extension in metres.

10. Two points, A and B, which are at different altitudes from planet Z, are shown in the diagram below.



The work done by the gravitational field to bring a spacecraft of mass 5000 kg from A to B is 5×10^{10} J. What is the gravitational potential at A if the gravitational potential at B is -5×10^7 J kg^{-1}?

A -6×10^7 J kg^{-1} B -5×10^7 J kg^{-1} C -4×10^7 J kg^{-1} D -1×10^7 J kg^{-1}

Answer: C

$$\text{Work done by the field} = m(V_A - V_B)$$

$$5 \times 10^{10} = 5000(V_A + 5 \times 10^7) \Longrightarrow V_A = -4 \times 10^7 \text{ J kg}^{-1}$$

Explanation: The field does positive work when the spacecraft moves to a lower potential (closer to the planet). So A, further out, has a higher (less negative) potential than B.

11. The atom of a monatomic ideal gas at absolute temperature T has an average kinetic energy E. What is the average kinetic energy of an atom of a diatomic ideal gas at temperature $2T$?

A $\frac{6}{5}E$ B $\frac{5}{3}E$ C $2E$ D $\frac{10}{3}E$

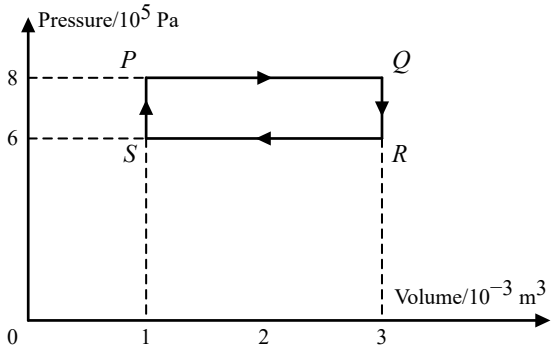
Answer: D

$$\text{Monatomic: } E = \frac{3}{2}kT$$

$$\text{Diatomic at } 2T\text{: } \frac{5}{2}k(2T) = 5kT = \frac{10}{3}E$$

Explanation: Each degree of freedom has $\frac{1}{2}kT$. A monatomic gas has 3 and a diatomic gas has 5 degrees of freedom, and the temperature is also doubled.

12. A gas of fixed mass which undergoes changes of pressure and volume is shown in the diagram below.



What is the work done by the gas during the expansion?

A 400 J B 1200 J C 1600 J D 2800 J

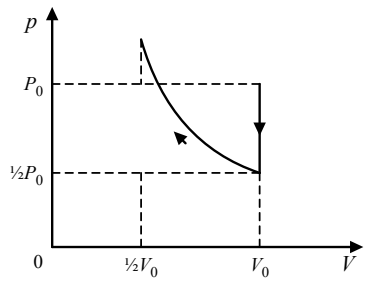
Answer: C

Expansion is $P \rightarrow Q$ at 8×10^5 Pa: $W = p\Delta V = (8 \times 10^5)(3 - 1) \times 10^{-3} = 1600$ J

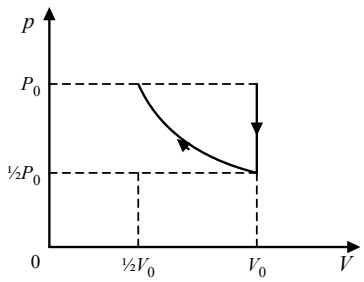
Explanation: Work is done by the gas only when it expands, along PQ. The net work for the whole cycle would be 400 J, but the question asks only for the expansion.

13. A monatomic ideal gas is initially at pressure P_0 and volume V_0. The gas is cooled at constant volume until its pressure is halved. The gas is then compressed adiabatically until its volume is halved. If the value of γ for the gas is $\frac{5}{3}$, which graph shows the variation of pressure, p, with volume, V, for the gas?

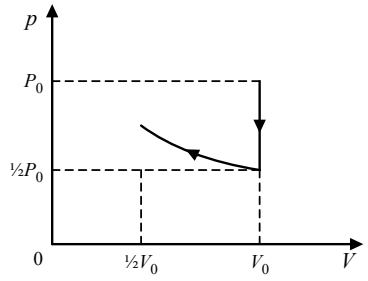
A



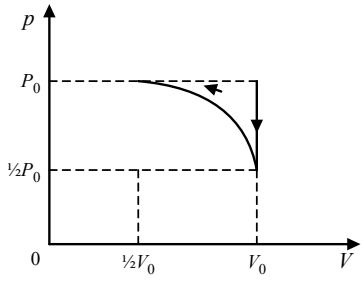
B



C



D



Answer: A

Step 1 (constant volume): vertical line at V_0 from P_0 down to $\frac{1}{2}P_0$.

Step 2 (adiabatic): $p = \frac{1}{2}P_0 \times 2^{5/3} = 1.59P_0$ at $\frac{1}{2}V_0$

The adiabatic curve rises steeply and ends above P_0.

Explanation: Use pV^{γ} = constant: halving the volume multiplies the pressure by $2^{5/3} = 3.17$, so the final pressure is greater than P_0. Only graph A ends above P_0.

14. Which method is **not** used to improve heat transfer by convection?

A A ventilation fan is fixed at the roof of a building.
B An air conditioner is fixed at a higher place in a room.
C The heating element in a kettle is fixed at the bottom of the kettle.
D A thin layer of air is formed between two glass planes in a double-glazed window.

Answer: D

A thin air layer in a double-glazed window is used to **prevent** convection (the air is too thin to circulate), not to improve it.

Explanation: A roof fan removes rising warm air, an air conditioner placed high lets cool air sink, and a kettle element at the bottom lets heated water rise: all of these use convection.

15. Which radiation is best absorbed by greenhouse gases which cause global warming?

A Infrared B Ultraviolet C Visible light D Radio waves

Answer: A

Greenhouse gases best absorb infrared radiation emitted by the warm Earth.

Explanation: The Earth's surface, at about 290 K, emits mainly infrared radiation. Greenhouse gases absorb it and re-emit some back towards the Earth, warming it.

Section B [15 marks]

16. (a) A satellite moves in a circular orbit at a speed of 4400 m s^{-1}. Determine the orbital period. [4]

Answer

Gravity provides the centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$ — 1

$r = \frac{GM}{v^2} = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})}{4400^2} = 2.06 \times 10^7$ m — 1

$T = \frac{2\pi r}{v} = \frac{2\pi(2.06 \times 10^7)}{4400}$ — 1

$T = 2.94 \times 10^4$ s — 1

4

Explanation: The satellite orbits the Earth, so use the mass of the Earth from the list of constants. First find the orbit radius from the speed, then the period.

(b) A planet has a radius of 5000 km. The acceleration of free fall of the planet is 6.05 m s^{-2}. Determine the

(i) mass of the planet, [2]

Answer

$g = \frac{GM}{R^2} \Rightarrow M = \frac{gR^2}{G} = \frac{(6.05)(5.00 \times 10^6)^2}{6.67 \times 10^{-11}}$ — 1

$M = 2.27 \times 10^{24}$ kg — 1

2

Explanation: Convert 5000 km to 5.00×10^6 m before squaring.

(ii) escape velocity of the planet. [2]

Answer

$v_e = \sqrt{(\frac{2GM}{R})} = \sqrt{(2gR)} = \sqrt{(2(6.05)(5.00 \times 10^6))}$ — 1

$v_e = 7.78 \times 10^3$ m s^{-1} — 1

2

Total for question **8**

Explanation: Since $GM = gR^2$, the escape velocity can be written $\sqrt{(2gR)}$. Use this planet's g, not the Earth's.

17. (a) What is the basic principle used in the first law of thermodynamics? [1]

Answer

The principle of conservation of energy — 1

1

Explanation: The first law, $Q = \Delta U + W$, states that heat supplied to a system is used to increase its internal energy and to do work.

(b) An insulated cylinder is fitted with a frictionless light piston which contains 0.22 mol of diatomic ideal gas. The gas expands from 4.0×10^{-3} m^3 to 4.2×10^{-3} m^3 with an initial pressure of 2.0×10^5 Pa.

(i) Name the thermodynamic process. [1]

Answer

Adiabatic expansion — 1

1

Explanation: The cylinder is insulated, so no heat enters or leaves the gas ($Q = 0$).

(ii) Calculate the final pressure of the gas. [3]

Answer

Diatomic gas: $\gamma = \frac{7}{5} = 1.4$ — 1

$p_1 V_1^{\gamma} = p_2 V_2^{\gamma} \Rightarrow p_2 = (2.0 \times 10^5)(\frac{4.0}{4.2})^{1.4}$ — 1

$p_2 = 1.87 \times 10^5$ Pa — 1

3

Explanation: Use the adiabatic relation, not Boyle's law: the temperature falls during an adiabatic expansion.

(iii) Calculate the work done by the gas. [2]

Answer

$W = \frac{p_1 V_1 - p_2 V_2}{\gamma - 1} = \frac{(2.0 \times 10^5)(4.0 \times 10^{-3}) - (1.868 \times 10^5)(4.2 \times 10^{-3})}{0.4}$ — 1

$W = 38.7$ J — 1

2

Total for question **7**

Explanation: Equivalently, $W = -\Delta U = -nC_{V,m}\Delta T$, with T found from $pV = nRT$ (437.6 K falling to 429.1 K). The answer is only slightly less than $p_1\Delta V = 40$ J because the volume change is small.

Section C [30 marks]

18. (a)

(i) Define *centre of mass* of a system. [1]

Answer

The centre of mass is the point at which the whole mass of the system can be considered to be concentrated — 1

1

Explanation: A resultant force through the centre of mass gives the system linear acceleration without rotation.

(ii) A system consists of two point masses of 1.50 kg and 3.50 kg, separated by a distance of 0.55 m. Determine the centre of mass of the system. [2]

Answer

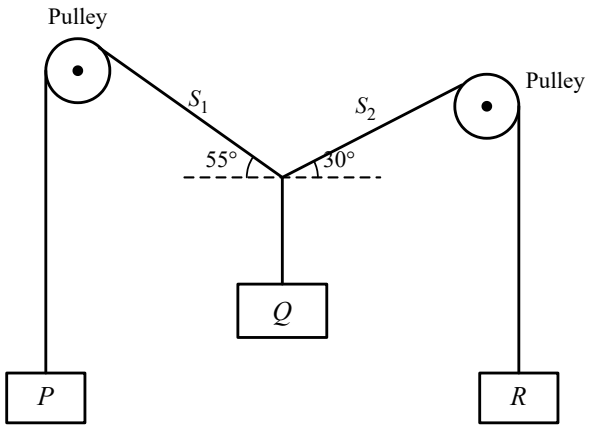
Taking the 1.50 kg mass as origin: $x = \dfrac{(1.50)(0) + (3.50)(0.55)}{1.50 + 3.50}$ — 1

$x = 0.385$ m from the 1.50 kg mass (0.165 m from the 3.50 kg mass) — 1

2

Explanation: State the reference point clearly. The centre of mass is nearer to the heavier mass.

(b) Three bodies, P, Q and R, are connected to each other by a string which passes through two smooth pulleys as shown in the diagram below.



The mass of Q is 5.0 kg while the strings, S_1 and S_2, are at 55° and 30° respectively with the horizontal. If the system is in equilibrium, determine the tension in S_1 and S_2. [4]

Answer

Horizontal: $T_1 \cos 55° = T_2 \cos 30°$	1
Vertical: $T_1 \sin 55° + T_2 \sin 30° = (5.0)(9.81) = 49.05$ N	1
$T_1 = 1.51T_2 \Longrightarrow 1.737T_2 = 49.05 \Longrightarrow T_2 = 28.2$ N (tension in S_2)	1
$T_1 = 42.6$ N (tension in S_1)	1
	4

Explanation: Consider the equilibrium of the knot above Q. The horizontal components of the two tensions balance, and their vertical components support the weight of Q.

(c) A uniform ladder of length 4.0 m and mass 20.0 kg leans against a vertical concrete wall at an angle of 40° with the concrete ground. The coefficients of friction at the floor and at the wall are the same, which is 0.45. A man of mass 70.0 kg climbs up the ladder and reaches point X on the ladder before the ladder starts to slip.

(i) Determine the normal reactions at the wall and at the floor. [5]

Answer

Forces: R_f up and friction μR_f towards the wall at the floor; R_w horizontal and friction μR_w upward at the wall	1
Horizontal: $R_w = 0.45R_f$	1
Vertical: $R_f + 0.45R_w = (20.0 + 70.0)(9.81) = 882.9$ N	1
$R_f(1 + 0.45^2) = 882.9 \Longrightarrow R_f = 734$ N	1
$R_w = 330$ N	1
	5

Explanation: At the point of slipping, friction at both surfaces is limiting. The foot tends to slide away from the wall, so floor friction points towards the wall; the top tends to slide down, so wall friction points up.

(ii) Calculate the position of point X on the ladder. [2]

Answer

Moments about the foot: $R_w(4.0 \sin 40°) + 0.45R_w(4.0 \cos 40°) = (20.0)(9.81)(2.0 \cos 40°) + (70.0)(9.81)d \cos 40°$	1
$1305 = 300.6 + 526.0d \Longrightarrow d = 1.9$ m from the bottom of the ladder	1
	2

Explanation: Taking moments about the foot removes both forces at the floor. The perpendicular distances use sin 40° for horizontal forces and cos 40° for vertical forces.

(iii) Between a rough floor and a rough wall, which will make the situation above safer? [1]

Answer

A rough floor 1

1

Total for question **15**

Explanation: Floor friction stops the foot of the ladder sliding out; without wall friction the ladder can still be in equilibrium, but without floor friction it cannot.

19. (a) The formula for the velocity, v, of longitudinal waves which travel along a rod made of material of Young's modulus E and density ρ is $v = \sqrt{(E/\rho)}$. Show that the formula is dimensionally correct. [3]

Answer

$[E] = [\text{stress}] = ML^{-1}T^{-2}$; $[\rho] = ML^{-3}$ 1

$[\sqrt{(\frac{E}{\rho})}] = \sqrt{(\frac{ML^{-1}T^{-2}}{ML^{-3}})} = \sqrt{(L^2T^{-2})} = LT^{-1}$ 1

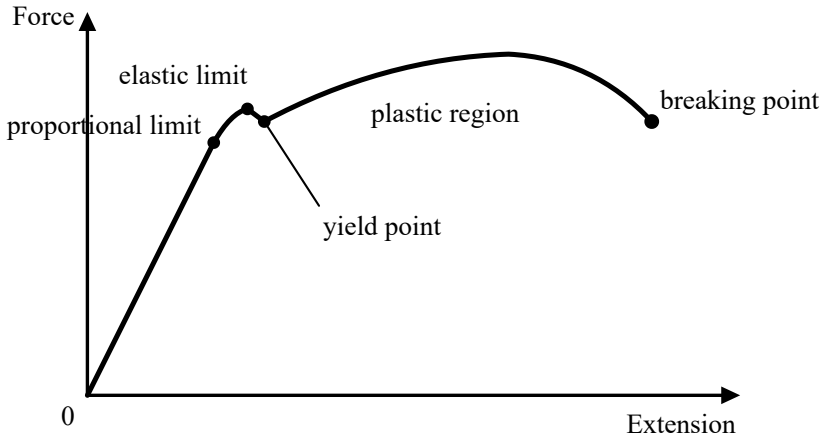
$[v] = LT^{-1}$; both sides have the same dimensions, so the formula is dimensionally correct 1

3

Explanation: Young's modulus has the same dimensions as stress, because strain is dimensionless.

(b) Sketch a graph of tensile force against extension of a copper wire until it breaks. Mark the yield point on the graph. [2]

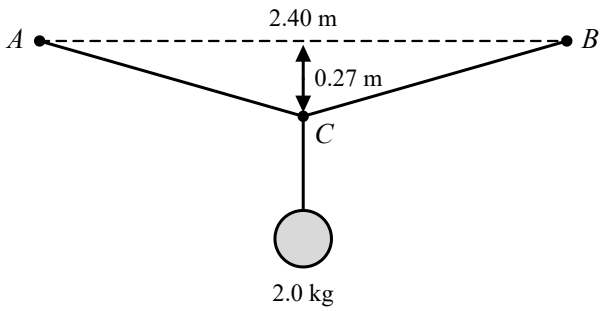
Answer



Correct shape: straight line, then a long plastic region up to the breaking point (ductile)	1
Yield point marked just after the elastic limit	1
	2

Explanation: Copper is ductile, so it shows a large plastic deformation. At the yield point the wire suddenly extends with little increase in force.

(c) A uniform wire of original length 2.40 m is connected to two fixed points, *A* and *B*, at a horizontal distance of 2.40 m apart as shown in the diagram below.



A mass of 2.0 kg is hung at the midpoint, C, of the wire. The equilibrium position of C is 0.27 m below the level of AB. Neglect the weight of the wire and take the Young's modulus for its material as 2.0×10^{11} N m^{-2}.

(i) Calculate the strain of the wire. [3]

Answer

Stretched length of half the wire = $\sqrt{(1.20^2 + 0.27^2)} = 1.23$ m — 1

Extension of half the wire = $1.23 - 1.20 = 0.03$ m — 1

Strain = $\frac{0.03}{1.20} = 2.5 \times 10^{-2}$ — 1

3

Explanation: Each half of the wire, originally 1.20 m, becomes the hypotenuse of a right-angled triangle with sides 1.20 m and 0.27 m. The strain is the same for the whole wire.

(ii) Calculate the stress in the wire. [2]

Answer

Stress = $E \times$ strain = $(2.0 \times 10^{11})(2.5 \times 10^{-2})$ — 1

Stress = 5.0×10^9 N m^{-2} — 1

2

Explanation: Young's modulus = stress/strain, so stress = $E \times$ strain.

(iii) Calculate the energy stored in the wire. [3]

Answer

Vertical equilibrium at C: $2T \sin \alpha = mg$, $\sin \alpha = \frac{0.27}{1.23} \Rightarrow T = \frac{(2.0)(9.81)}{2(0.2195)} = 44.7$ N — 1

Energy stored = $\frac{1}{2}Te$, total extension $e = 2(0.03) = 0.06$ m — 1

Energy = $\frac{1}{2}(44.7)(0.06) = 1.34$ J — 1

3

Explanation: The tension is the same throughout the wire. Only the vertical components of the tension support the 2.0 kg mass, and the energy stored is $\frac{1}{2} \times$ tension $\times$ total extension.

(iv) State **two** assumptions which you make in the calculation above. [2]

Answer

• The wire obeys Hooke's law (the elastic limit is not exceeded)	1
• The cross-sectional area of the wire remains constant / the wire is uniform	1
[Also accept: the weight of the wire is negligible; the supports are rigid]	
	2
Total for question	**15**

Explanation: The formulas used (E = stress/strain and energy = $\frac{1}{2}Fe$) are valid only within the proportional limit.

20. (a) State **two** assumptions of the kinetic theory for an ideal gas. [2]

Answer

• A gas consists of a very large number of molecules in continuous random motion	1
• The volume of the molecules is negligible compared with the volume of the container	1
• Collisions are perfectly elastic	1
• No intermolecular forces except during collisions	1
[Any two – maximum 2 marks]	
	2

Explanation: Use the word molecules, not particles.

(b)

(i) What is meant by *the internal energy of an ideal gas*? [1]

Answer

The internal energy of an ideal gas is the total kinetic energy of all its molecules (there is no potential energy because there are no intermolecular forces)	1
	1

Explanation: For a real gas the internal energy also includes the potential energy of the molecules due to intermolecular forces.

(ii) State **two** factors which affect the internal energy of an ideal gas. [2]

Answer

• Temperature of the gas	1
• Number of moles (molecules) of the gas / degrees of freedom of the molecules	1
	2

Explanation: $U = \frac{f}{2}nRT$: it depends on T, n and the number of degrees of freedom f, but not on the pressure or volume separately.

(c) The pressure, p, exerted by an ideal gas is given by $p = \frac{1}{3}\rho\langle c^2\rangle$. (Corrected by MPM erratum.)

(i) What is meant by $\langle c^2\rangle$? [1]

Answer

$\langle c^2\rangle$ is the mean square speed: the average of the squares of the speeds of all the molecules 1

1

Explanation: $\langle c^2\rangle = \frac{c_1^2 + c_2^2 + \ldots + c_N^2}{N}$. Its square root is the r.m.s. speed.

(ii) Use the equation above to derive an expression for the internal energy of an ideal gas in terms of the pressure and the volume of the gas. [4]

Answer

$\rho = \frac{Nm}{V} \Longrightarrow p = \frac{1}{3}\frac{Nm}{V}\langle c^2\rangle$ 1

$pV = \frac{2}{3}N(\frac{1}{2}m\langle c^2\rangle)$ 1

Internal energy of an ideal (monatomic) gas = total kinetic energy: $U = N(\frac{1}{2}m\langle c^2\rangle)$ 1

$U = \frac{3}{2}pV$ 1

4

Explanation: The kinetic theory equation contains the total translational kinetic energy of the molecules, which is the internal energy of a monatomic ideal gas.

(iii) Calculate the increase in the internal energy of an ideal gas when it expands from volume 2.1×10^{-3} m^3 to 6.6×10^{-3} m^3 at constant pressure of 100 kPa. [2]

Answer

$\Delta U = \frac{3}{2}p\Delta V = \frac{3}{2}(100 \times 10^3)(6.6 - 2.1) \times 10^{-3}$ 1

$\Delta U = 675$ J 1

2

Explanation: At constant pressure, the change in $\frac{3}{2}pV$ is $\frac{3}{2}p\Delta V$.

(iv) Determine the amount of heat to be supplied in the expansion above. [3]

Answer

Work done by the gas $W = p\Delta V = (100 \times 10^3)(4.5 \times 10^{-3}) = 450$ J	1
$Q = \Delta U + W = 675 + 450$	1
$Q = 1125$ J	1
	3
Total for question	**15**

Explanation: The heat supplied must both increase the internal energy and provide the work done as the gas expands against the constant pressure.